



**Verified Carbon
Standard**

DONGXING WASTE INCINERATION PLANT LEACHATE BIOGAS HARMLESS TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT



Document Prepared By LGAI Technological Center, S.A. (Applus+
Certification)

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Summary:

LGAI Technological Center, S.A. (hereafter referred to as “Applus+ Certification”) has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the verification of greenhouse gas emission reductions of the project activity “Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project” (VCS Ref. No. 3851, hereafter referred to as “the project activity”) reported in the monitoring report /1/ during monitoring period 21/01/2022 to 31/12/2023.

The project activity has been validated by Applus+ Certification based on the VCS PD /3/ version 04 dated 25/06/2023 and reported in the validation report No. A+SH_SYST_VCS_VAL_11522 /4/, version 01.3, completed on 26/06/2023 which is available at <https://registry.terra.org/app/projectDetail/VCS/3851>.

The project activity is a biogas recovery and power generation project located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China, of which the main purpose is to use biogas for electricity generation. The total installed capacity of the project activity is 3 MW consisting of 3 sets of 1,000 kW generators. The project uses biogas generated by the anaerobic treatment section of leachate for power generation. During the fixed 10 years crediting period, the power generated by the project is expected to be 118,470 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the biogas recovered by project, the project activity avoids the emission of methane that would be generated

from the anaerobic treatment section of leachate. It's estimated that the project activity could achieve GHG emission reductions of 663,100 tCO₂e during the fixed crediting period.

The purpose and scope of this verification is to ensure that reported emission reductions are complete and accurate in accordance with applicable VCS standards and relevant UNFCCC requirements in order to be certified. A desk review and a site visit have been conducted to verify the data submitted in the monitoring report /1/. Applus+ Certification confirms the following has been reviewed:

- Monitoring plan included in the registered VCS PD /3/ version 3.0 dated 25/06/2023;
- Validation report No. A+SH_SYST_VCS_VAL_11522 /4/ version 01.3, dated 26/06/2023;
- AMS-III.H: Methane recovery in wastewater treatment, version 19.0 /7/;
- AMS-I.D: Grid connected renewable electricity generation, version 18.0 /7/;
- VCS standards version 4.7 and VCS program guide version 4.4, as well as relevant UNFCCC requirements;
- All information and references relevant to the project activity's resulting in emission reductions.

During this verification, no finding was identified related to the monitoring, implementation or operations of the project activity in relation to relevant VCS standards, guidance and UNFCCC requirements and relevant host party criteria and the applied baseline and monitoring methodology etc.

Applus+ Certification confirms that the project is implemented in accordance with the registered PD /3/. The monitoring plan complies with the applied methodology AMS-III.H version 19.0, AMS-I.D version 18.0 /7/ and the monitoring has been carried out in accordance with the registered PD. The monitoring system is in place and the emission reductions are calculated without material misstatements. The level of assurance of the verification is reasonable. Our opinion relates to the projects GHG emissions and the resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring and its associated documents. Based on the information reviewed and evaluated Applus+ Certification confirms that the implementation of the project has resulted in 123,622 tCO₂e emission reductions during period 21/01/2022 to 31/12/2023.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center, S.A. (Applus+ Certification) has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the verification of greenhouse gas emission reductions of the project activity “Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project” (VCS Ref. No. 3851) reported in the Monitoring Report /1/ during monitoring period 21/01/2022 to 31/12/2023.

LGAI Technological Center, S.A. (Applus+ Certification) as the verification body of the project activity has been accredited as a DOE by UNFCCC and also meets the competence requirements as set out in ISO 14065:2020.

The objective of verification is to have an independent review and ex post determination by a Validation and Verification Body (VVB) of the monitored reductions in GHG emissions that have occurred as a result of the registration of VCS project. Certification is the written assurance by the VVB that, during a specific time period, a proposed VCS project activity achieved the reductions in anthropogenic emissions by sources of GHGs as verified.

The objective of this verification/certification is to verify and certify emission reductions, reported for the “Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project” in China for the period 21/01/2022 to 31/12/2023.

1.2 Scope and Criteria

The verification scope is defined as an independent and objective review of the registered PD, the Project’s baseline study and Monitoring Report (MR) and other relevant documents. The information in these documents is reviewed against VCS Version 4.7 requirements, UNFCCC rules and associated interpretations.

The verification has been conducted in accordance with the VCS Standard version 4.7, other valid and applicable VCS Regulatory Documents and templates (see Appendix 2 of this verification Report below), UNFCCC rules and requirements (where applicable) and associated decisions of the VCS Secretariat and the UNFCCC Executive Board (where applicable) (hereinafter also can be referred to as regulatory documents).

The verification is not meant to provide any consulting towards the client. However, stated requests for forward actions and/or corrective actions may provide input for improvement of the Project monitoring towards reductions in the GHG emissions.

1.3 Level of Assurance

The verification report is based on the VCS-PD, VCS Monitoring Report (MR), supporting evidences made available to the verifier as well as other supporting evidences made available to the VVB and information collected through performing interviews and during the on-site assessment.

Through an exhaustive process of documentary review, interviews with the project owners, site visits and sampling and surveying (where applicable), assessment on GHG Emission Reductions calculations against applicable methodology and any other regulatory document, and reporting of objective evidences, the VVB is able to reach a reasonable level of assurance on the conduction of the Verification process.

The achieved level of assurance for this verification is thus a reasonable level of assurance as per the requirements set out in the applied VCS Standard version 4.7 clause 4.1.10 (1).

VVB applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per VCS Standard version 4.7 clause 4.1.10 (4).

Additionally, according to the VCS Validation and Verification Manual version 3.2, while all material errors, omissions and misrepresentations must be addressed for a project to receive a positive validation or verification opinion, when non-material errors have been found in the project documents, the VVB requested the project proponent to correct them and ensured that such errors have been addressed where practicable.

The verification conclusion is assured a reasonable level of assurance.

1.4 Summary Description of the Project

Project title	Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project
VCS reference number	3851
Project Proponents	Henan BCCY Environmental Energy Co., Ltd. (Project Owner, host country, P. R. China)
Location of the project	Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China. Geographic coordinates: longitude of 113.57° E and latitude of 34.37° N
Project start date	Construction start date: 15/09/2021

	Operation start date: 21/01/2022
Version of PD	VCS PD Version 04, dated 25/06/2023
Monitoring period	21/01/2022 to 31/12/2023
First monitoring report	Version 01, dated 22/03/2024
Final monitoring report	Version 03, dated 19/06/2024
Applied Methodology/Version	AMS-III.H, version 19.0 AMS-I.D, version 18.0
Scope/Technical Area	1/1.1 and 13/13.1 Scope 01: energy industries (renewable-/non-renewable sources) and sectoral scope 13: Waste handling and disposal

The project activity is a biogas recovery and power generation project located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China, of which the main purpose is to use biogas for electricity generation. The total installed capacity of the project activity is 3 MW consisting of 3 sets of 1,000 kW generators. The project uses biogas generated by the anaerobic treatment section of leachate for power generation. During the fixed 10 years crediting period, the power generated by the project is expected to be 118,470 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the biogas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 663,100 tCO₂e during the fixed crediting period.

The project activity has been validated by Applus+ Certification based on the VCS PD /3/ version 04 dated 25/06/2023 and reported in the validation report No. A+SH_SYST_VCS_VAL_11522 /4/, version 01.3, completed on 26/06/2023 which is available at <https://registry.verra.org/app/projectDetail/VCS/3851>.

2 VERIFICATION PROCESS

2.1 Method and Criteria

Verification was conducted using Applus+ Certification's procedures in line with the requirements specified in the VCS Standard version 4.7, CDM M&P, the latest version of the CDM Validation and Verification Standard, and relevant UNFCCC requirements and applying standard auditing techniques.

Applus+ Certification verified the implementation of the monitoring plan and the data presented in the Monitoring Report for the period in question. This involved a site visit and a desk review of the Monitoring Report. This Verification Report describes the findings of this assessment.

The Verification process has been conducted following the regulatory documents and based on the ISO 14064-3:2019 and ISO 14065:2020 using standard auditing techniques as required by the VCS Standard version 4.7.

The Verification process has been conducted through an onsite inspection (see below), thus, no need for a risk analysis for remote inspections has been determined for this Validation (cf. VCS Standard version 4.7. Para 4.1.13).

No ICT means have been used for the conduction of this Verification process other than those for the provision of documents and transferring information and undertake normal communications with the PP(s). Those ICT means supporting the process in this sense, have been used to an extent sufficient to cover the background necessities of the Verification process and have demonstrated enough effectiveness to conduct it in accordance with the relevant applicable regulatory requirements.

This process has included: the review of data and information presented to verify their completeness; the review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements and the QA/QC procedures; an evaluation of data management; and an evaluation of the QA/QC system in place in the context of its influence on the generation and reporting of emission reductions.

No virtual sites have been included in the Verification process according to the definition in the IAF MD4: 2023 document. Instead, the documentation has been provided upon request from the VVB by other means (such as physical documents, mail, transfer applications, etc.)

Applus+ Certification has completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project's emissions and emission reductions, including leakage if relevant;
- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;
- Controls on the collection and handling of data;

- Means of verifying reported data; and
- Compilation of the monitoring report.

Based on the above, the VVB prepared a Verification Plan and applied an evidence-gathering plan designed to collect sufficient and appropriate evidence upon which to base the conclusion and to consider the inherent risk for the verification. The evidence-gathering plan is based on analytical procedures and tests and considers the thresholds for materiality as per the regulatory documents applied for this Verification (5% as per the applied VCS Standard version 4.7).

The applied evidence-gathering plan is used by the VVB to determine the conformity of the statements made by the client against the principles and requirements of the applied regulatory documents during this verification.

The evidence-gathering plan designed by the VVB takes into consideration, as a minimum:

- The selection and management of the data and information;
- The processes for collecting, processing, consolidating, aggregating and reporting the data and information;
- Systems and processes in place to ensure the validity and accuracy of the data and information;
- The design and maintenance of the system to control the data and information;
- Systems, processes and personnel that support system in place, including activities for ensuring data quality;
- The plans for instrument maintenance and calibration where appropriate;
- The results of previous validation activities and verifications, where available and appropriate

Applus+ Certification has verified the implementation of the monitoring plan and the data presented in the Monitoring Report (MR) version 03, dated on 19/06/2024 for the monitoring period applicable to this verification process.

The process of verification involved the following relevant milestones:

- The signature of the contract for verification between the PP and the VVB;
- A kick-off meeting/communication to require the documentary material to perform an initial desk review;
- An initial desk review of the documentary evidences prepared by the Project Proponent;
- The issuance of findings coming from this initial desk review (if any);
- The conduction of an onsite audit, including interviews and video-calls in substitution of a site visit (see Sections 2.3 and 2.4 below);
- The issuance of findings from the onsite audit and communication of deviations (if any) to the Project Proponent;
- The assessment of their resolution and closure, when appropriate;
- Once all the findings have been closed, the elaboration of this Verification Report, in which such findings (if any) are described;
- The conduction of an independent Technical Review;

- Upon conclusion of the Technical Review, the VVB may perform a final quality check before the issuance of the Final Verification Report (FVR) and signature of the corresponding Verification Deed.

The process of this verification assessment has not implied any sampling plan.

The information of the assessment team is included in below:

Assessment team

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of assessment team has to be approved by the LGAI Technological Center S.A. (Applus+ Certification) Central Site during its Contract Review stage, ensuring that the required skills are covered by the team.

The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A).
- Technical Expert (TE).
- Technical Reviewer (TR).
- Other supporting roles such as Country Expert (CE) or Financial Expert (FE)
- Any of the above-mentioned roles in training (iT, e.g. AiT for auditor in training).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the Assessment Team as shown below:

Name	Role	SS/TA Knowledge	Financial Expertise	Host country experience	Attendance to on-site visit
Doris Dai	LA/TE	Yes (1.1 and 13.1)	n/a	Yes	Yes
Simon Shen	TR /TE	Yes (1.1 and 13.1)	n/a	Yes	No

Doris Dai (Master's Degree in Environmental Sciences, Bachelor's Degree in Environmental Technology) is an Auditor appointed by Applus+ Certification (LGA Technological Center, S.A) for the GHG project assessment and auditing. She has more than 8 years of work experience in CDM/VCS project assessment and more than 3 years of working experience in GS4GG project assessment. Before she joined Applus+

Certification (LGAI Technological Center, S.A), she has been working for CTI Certification as senior GHG Auditor for 3.5 years.

Simon Shen (Master's degree in Thermal Energy Engineering, Bachelor's Degree in Environmental Engineering) is an Auditor appointed by Applus+ Certification (LGAI Technological Center, S.A) for the GHG project assessment, auditing and technical review. He has more than 8 years of work experience in CDM/GS4GG/VCS project assessment and review with Applus+, apart from the years of experience working as GHG Auditor and ISO 9001/14001 in TUV SUD for 3.5 years before he joined Applus+.

Mr. Simon Shen has extensive experience also as former Applus+ Shanghai CDM Technical Manager. Since 2014, Mr. Simon Shen works as an external individual in Applus+. At the time, he participated in plenty of Chinese CCER audits and enterprises carbon emissions verifications.

2.2 Document Review

The VCS monitoring report /1/ version 01 dated 22/03/2024, version 03 dated 19/06/2024 and the emission reduction calculations spreadsheet /2/, were assessed as part of the verification. In addition, the VCS PD /3/ version 04 dated 25/06/2023 in particular the baseline estimations and the monitoring plan, the VCS Validation Report /4/ dated 26/06/2023, as well as relevant documents, were reviewed. A detailed document reviewed are listed in Appendix 2 of the report.

2.3 Interviews

A physical site visit interview was made by the assessment team on 08-09/04/2024, The key personnel interviewed through the sit visit are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Tao Jinlong	General Manager	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance.
Mr. Li Wei	Operator	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	
Mr. Liu Shaojun	Operator	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	
Mr. Xu Puzhou	Operator	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	
Mr. Liu Shaojun	Villager	Zhongmu County	
Ms. Tian Qiming	Villager	Zhongmu County	
Ms. Wang Lei	Project Manager	Henan BCCY Environmental Energy Co., Ltd.	Data collection and ER calculation.
Ms. Xu Yaya	Project Manager	Henan BCCY Environmental Energy Co., Ltd.	

2.4 Site Visits

The assessment team performed the on-site verification (Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China) on 08-09/04/2024. The interviewed personnel including the role and organization and objective are listed in above table in the section 2.3, please refer to above for details.

2.5 Resolution of Findings

As an outcome of the verification process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the Project Proponents, or if the evidence provided to prove conformity is insufficient;
- b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the Project Proponents;
- c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the Project Proponents.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

All CARs and CLs raised during verification shall be resolved prior to submitting a request for issuance.

There is no CARs and CLs raised for this monitoring period for the project.

2.5.1 Forward Action Requests

None FAR was raised during the verification process. Also, there are no remaining from former validation.

2.6 Eligibility for Validation Activities

Not applicable as LGAI Technological Center, S.A. holds the accreditation for the validation and verification for projects under scope 1 and 13.

3 VALIDATION FINDINGS

3.1 Methodology Deviations

Not applicable as no deviation for methodology.

3.2 Project Description Deviations

Not applicable as no deviation for project description.

3.3 New Project Activity Instances in Grouped Projects

Not applicable as not a grouped project.

3.4 Baseline Reassessment

Not applicable as the project does not undergo baseline reassessment during the monitoring period.

4 VERIFICATION FINDINGS

4.1 Project Details

Item	Evidence gathering activities, evidence checked, and assessment conclusion:
Audit history	By checking registered PD /3/ and information on the VERRA website, the audit history mentioned in the MR has been confirmed to be correct
Double counting and participation under other GHG programs	Through reviewing the registered PD /3/ and validation report /4/, it was validated that the project has been registered as a VCS project with reference No. 3851 which is available at https://registry.verra.org/app/projectDetail/VCS/3851. Sufficient information has been provided for registered or seeking registration under any other GHG programs or rejected by another GHG program. The project does not participate in other emissions trading program by checking public information on Internet, interviewing with project owner and statement issued by project owner. During the period from 21/01/2022 to 20/01/2032 as the VCS crediting period, the project would claim only for VCU. Therefore, Applus+ Certification consider the project is eligible to participate under the VCS Program as there is no double counting for the emission reduction during any period.
No double claiming with emissions trading programs or binding emission limits	The GHG emission reductions or removals generated by the project have not become included in an emissions trading program or any other mechanism that includes GHG allowance trading. The same has been checked through declaration from the project proponents. The assessment team checked the REC Mechanism database of China, Chinese Emission Trading System (Chinese ETS) and Chinese Certified Emission Reduction Mechanism (CCER), and found that the project activity is not accredited / registered under REC mechanism, Chinese ETS, or CCER mechanism. Therefore, Applus+ Certification confirm no double claiming with emissions trading programs or binding emission limits.

No double claiming with other forms of environmental credit	<i>The project has not received or sought any other form of environmental credit or has become eligible to do so since validation or previous verification. The same has been checked through declaration from the project proponents.</i> <i>By checking information on I-REC website /20/ and REC Mechanism database of China, it is confirmed that the project does not apply for REC of China or I-REC.</i> <i>Therefore, Applus+ Certification confirm no double claiming with other forms of environmental credit.</i>
Supply chain (scope 3) emissions double claiming	<i>The project does not involve in supply chain (scope 3) emissions double claiming as all electricity generated by the project was supplied to power grid.</i> <i>The project is biogas power generation project does not involve in supply chain (Scope 3).</i>
Sustainable development contributions	<i>By checking registered VCS PD /3/, EIA Report /28/, HR Records /16/, China's National Plan on Implementation of the 2030 Agenda for Sustainable Development and 17 SDGs defined by UNDP, interviewing with stakeholders during site visit, the verification team confirmed that the project activity would contribute sustainable development in the region and China's Sustainable Development Goals (SDG). The specific sustainable development goals that the project activity could achieve have been listed as follows:</i> SDG 13 Climate Action <i>The project recovers and destroys biogas, mainly methane, which would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030. Total GHG emission reductions and removals of 663,100 tCO₂e will be achieved during the fixed 10 years crediting period.</i> SDG 8 Decent works and Economic Growth <i>The project can provide permanent and full-time job opportunities over the project lifetime.</i> SDG 7 Affordable and Clean Energy

	The project utilizes the leachate biogas from Zhengzhou (East) environmental Protection Energy Project to generate electricity by gas engines. Total electricity generation supplied by the project to the grid is estimated to be 118,470 MWh, which could replace the equivalent electricity from fossil fuel based grid. Moreover, by checking public information, staff roaster of project /16/ and site visit, information mentioned in the Table of Sustainable Development Contributions in the MR has been confirmed to be correct.
Additional information relevant to the project	No commercially sensitive information has been excluded from the public version of the PD or MR. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this verification.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	By checking registered PD /3/, and site visit, the stakeholder involved in the project has been correctly identified in the MR and in line with the same in the PD as below: The project is located in Zhengang Town, Zhongmu County, Zhengzhou City, Henan Province, P. R. China, local residents in Zhengang Town are considered as stakeholders. Since the project is to collect the leachate biogas from Zhengzhou (East) environmental Protection Energy Project and utilize it to generate electricity, employees of Zhengzhou (East) environmental Protection Energy Project are considered as stakeholders. The assessment team considers the stakeholder identification of the project activity as reasonable.
Legal or customary tenure/access rights	By checking Cooperation agreement signed by the project owner and Zhengzhou Dongxing Environmental Protection Energy Co., Ltd /21/, it is confirmed that the project owner has the legal right to use the biogas generated from Zhengzhou (East) environmental Protection Energy Project for power generation.

	By site visit, it is confirmed the project does not involved in land issues. Therefore, it was verified the project activity does not violate any legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.
Stakeholder diversity and changes over time	Consider there are no change for the project since the validation and the project is implemented as the design mentioned in the PD, the stakeholder diversity and changes over time is highly unlikely and this is the same as actual scenario as no stakeholder diversity and changes occurs. And during this monitoring period, there are no change for the stakeholders.
Expected changes in well-being	Following expected changes in well-being have been identified and confirmed to be correct through the site visit: Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing biogas to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 118,470 MWh, which could replace the equivalent electricity from fossil fuel based grid.
Location of stakeholders	By site visit, it is confirmed that the location of stakeholders is in Zhongmu County, Zhengzhou City, Henan Province, P. R. China.
Location of resources	By site visit, it is confirmed that the location of resources is in Zhongmu County, Zhengzhou City, Henan Province, P. R. China.

4.2.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Ongoing consultation	By site visit interview, it is confirmed that during the monitoring period, a grievance expression book /22/ was put in the office of the project,

	which is used to collect stakeholders' views about the project. Project Proponent checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Meanwhile, during this monitoring period, stakeholder meetings has been held.
Date(s) of stakeholder consultation	By checking questionnaires /23/, it is confirmed that a local stakeholder consultation meeting was held on 25/11/2021 by the project owner. Before the stakeholder consultation meeting, the project information with contact information and the stakeholder meeting notice were published on the bulletins at and nearby the project site on 10/11/2021 to invite Local stakeholders. During the meeting, the project information and the continuous input / grievance mechanism have been introduced to the local stakeholders. The project information with contact information has also been posted on the bulletins at and nearby the project site. During the meeting, a survey was carried out on the local residents and comments received from the survey are summarized as follows. The survey was conducted through distributing and collecting responses to a questionnaire. In total 30 out of 30 questionnaires were returned with a 100% response rate. By checking meeting minutes /24/ and site visit, it is confirmed that during this monitoring period, 2 meetings were organized at Dongxing Waste Incineration Plant on 16/08/2022 and 22/06/2023 to collect comments from all stakeholders, such as representatives of local residents, Zhengzhou (East) environmental Protection Energy Project, and the project owner and so on. The project owner appointed one person to make a meeting minute.
Communication of monitored results	By checking the questionnaires, it is confirmed that the local stakeholder has no negative comments for the construction of the project activity. By checking meeting minutes and site visit, it is confirmed during this monitoring period, 2 stakeholder consultation meetings were held. At each meeting, the plant manager informed the stakeholders about the recent operation of the project. Questions raised by stakeholders were answered on site.

Consultation records	By checking the questionnaires, questionnaires have been filled out during the stakeholder meetings before the implementation of the project. By checking grievance expression book, meeting minutes and site visit, it is confirmed that each time a stakeholder consultation meeting is held, one person is designated to record the meeting. The questions and suggestions raised by stakeholders at the meeting were recorded. All the information has been recorded as meeting minutes. A grievance expression book was put in the office of the project, which is used to collect stakeholders' views about the project. Project Proponent checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information.
Stakeholder input	By checking questionnaires, grievance expression book, meeting minutes and site visit, it is confirmed that there were no negative inputs received during both construction and operation phase.

4.2.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	By checking Project approval /25/ and EIA approval /26/ issued by government, it is confirmed the project has been approved by Chinese government. By checking Cooperation agreement signed by the project owner and Zhengzhou Dongxing Environmental Protection Energy Co., Ltd, it is confirmed that the project owner has the legal right to use the biogas generated from Zhengzhou (East) environmental Protection Energy Project for power generation. By checking questionnaires, grievance expression book, meeting minutes and site visit, it is confirmed that local stakeholders held full right for FPIC. Therefore, it was verified the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.

	<i>And the project does not exacerbate nor influence the outcomes of unresolved conflicts.</i>
Outcome of FPIC discussion	<i>As mentioned above, the project has been approved by Chinese government and local management institute. And local stakeholders have no negative comments for the construction and operation of the project.</i> <i>The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.</i> <i>The assessment team confirmed that the local stakeholders held full right for FPIC. And the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.</i>

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	<i>Not applicable as no negative comments have been received for the project during both construction and operation phase of the project. Also, the assessment team confirm the procedure is easily accessible to stakeholders for ongoing consultation as described as above.</i>
Grievance redress procedure	<i>Not applicable as no negative comments have been received for the project during both construction and operation phase of the project. Also, the assessment team confirm the procedure is easily accessible to stakeholders for ongoing consultation as described as above.</i>

4.2.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
No negative comments received from public channel.	Not applicable as no negative comments received from public channel.	By checking questionnaires, grievance expression book, meeting minutes, VERRA website and site visit, it is confirmed that there were no negative inputs received during both construction and operation phase.

4.2.6 Risks to Local Stakeholders and the Environment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Risks to stakeholder participation	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. Also, during the site visit, no risks have been identified to stakeholder participation. Therefore, it was verified no risks have been identified to stakeholder participation.
Working conditions	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. Also, during the site visit, it is confirmed that the staff of the project is working under indoor environment and safety protection equipment have been provided in case for on-site monitoring and device maintenance. No risks identified for working conditions.

Safety of women and girls	<i>By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws /27/ set up by Chinese government in aspect of environment, health and safety. Moreover, China has published "Women's Rights Protection Law" to provide support and protection to women and girls.</i> <i>In order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations", prohibiting employers from recruiting minors under the age of 16. Also, China published "Women's Rights Protection Law". Therefore, the Project will not expose women and girls to further risks or hazards. By checking HR Records and Staff Roaster, it was verified no girls have been recruited, also equal payment has been made to women and men.</i> <i>Also, during the site visit, it is confirmed that no risks has been identified to safety of women and girls.</i>
Safety of minority and marginalized groups, including children	<i>By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of minor protection.</i> <i>The country has strict prohibition for child labour. Thus, project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.</i> <i>Besides, in order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations", prohibiting employers from recruiting minors under the age of 16. By checking HR Records and Staff Roaster and site interview, the assessment team was able to confirm that no Minors under the age of 16 were employed.</i> <i>Also, during the site visit, it is confirmed that no minors have been hired by the project owner. No child labor detected for the project. No risks identified for safety of minority and marginalized groups, including children as not applicable for the project.</i>

Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)

By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction will be treated according to the measures in EIA report, and there will be no significant impact on the environment.

During the operation period, all the mitigation measures proposed by EIA report /28/ will be implemented and the following key aspects will be addressed:

Waste water

The wastewater produced by the project consists of biogas condensation waste water, circulating cooling waste water of generator set, and reclaimed waste water of soft water equipment. Condensation wastewater will be treated by leachate treatment system of Zhengzhou Dongxing Environmental Protection Energy Co., LTD., the sewage is reused and not discharged. Therefore, it can meet the standards of "The reuse of urban recycling water- Waste quality standard for industrial uses"(GB/T19923-2005).

Waste Gas

The exhaust gas produced by this project is mainly the exhaust gas of the generators. The concentration of the exhaust gas (mainly SO₂, NO_x and particulate matter) after combustion in gas-generator sets and processed by the SCR denitration system is discharged through the 15m high cylinder exhaust.

Noise pollution

According to the prediction results, the acoustic environment quality of the project equipment noise transmitted to the northwest boundary of project can meet the Class 2 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008). The acoustic environment quality of the project equipment noise transmitted to the southeast boundary of project can meet the Class 4 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008).

Solid waste

	The solid waste of the project is mainly sulfur paste, waste oil, waste oil barrel and waste denitration catalyst. The sulfur paste is stored in the sulfur storage room and sold regularly. The waste oil, waste oil barrels and waste denitration catalyst are stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The construction standard of hazardous waste room shall meet the requirements of "Standard for pollution control on hazardous waste storage" (GB18597-2001). During this monitoring period, by checking test report issued by Henan Provincial Government Testing Co., LTD, Hazardous waste collection and disposal contract with Henan Junchi Environmental Protection Technology Co., LTD. and site visit, it is able to confirm measurement has been implemented to control the potential negative environmental impacts.
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4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination and sexual harassment	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena. By site visit, it is confirmed no discrimination and sexual harassment detected. The project does not create any direct or indirect impacts on gender equality and/or the situation of women:

	a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls. b. The Project will not restrict women's rights or access regarding natural resources. c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.
Management experience	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government which means the government recognize the experience of the project owner for project construction and operation. Moreover, the project owner of the project has developed several biogas projects and therefore has extensive development and management experience.
Gender equity in labor and work	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena. By site visit, it is confirmed no violation for Gender equity in labor and work detected. The project, complying with the "Women's Rights Protection Law" and "Labor Contract Law" does not discriminate the local community on basis of gender or caste or religion and therefore equally serve to all. By checking HR Records and interviewing staff from the project owner, it was verified the Project design neither increase women's workload nor prevent them from engaging in other activities. The Project has equal opportunity for women and men to contribute both in volunteer and working positions. The project has HR policy and same is followed equally. The project owner has a specified HR policy that considers participation by both men and women.

	Besides, China also ratified relevant ILO core conventions on equality, namely Equal Remuneration Convention (Convention No 100). Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.
Human trafficking, forced labor, and child labor	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published “Labor Law” and “Criminal Law”, which could prohibit such phenomena. Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China. China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women’s Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee. China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview. By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.

4.2.7.2 Human Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Human rights	By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights.

	<i>China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties—the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law.</i> <i>Therefore, it is confirmed that the project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women.</i>
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4.2.7.3 Indigenous Peoples and Cultural Heritage

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	<i>By checking EIA report approved by government, it is confirmed that no indigenous peoples present in or within the area of influence of the project, the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.</i> <i>Hence not applicable for the project.</i>

4.2.7.4 Property Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Disputes over rights to territories and resources	<i>Not applicable as no IPs’, LCs’, and customary rights holders’ property rights was identified.</i>
Respect for property rights	<i>Not applicable as no IPs’, LCs’, and customary rights holders’ property rights was identified.</i>

4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Summary of the benefit sharing plan	<i>Not applicable as no property rights involved for the project.</i>
Benefit sharing during the monitoring period	<i>Not applicable as no property rights involved for the project.</i>

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	<i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>
Soil degradation and soil erosion	<i>The Project's purpose is to supply electricity from biogas. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.</i> <i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>
Water consumption and stress	<i>The Project's purpose is to supply electricity from biogas. Therefore, the impact of the Project on water consumption and stress is negligible.</i> <i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>
Usage of fertilizers	<i>The project is biogas recovery and utilization project. Therefore, the Project does not involve the application of fertilizers.</i> <i>By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.</i>

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species or habitat	<i>By checking EIA report approved by government, it is confirmed that the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.</i>

4.2.8.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	Not applicable as no rare, threatened, or endangered species involved.

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	Not applicable as no rare, threatened, or endangered species involved.

4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	Not applicable as the project is not a ARR, ALM, WRC or ACoGS project.

4.3 Accuracy of Reduction and Removal Calculations

By means of on-site visit, the assessment team confirms that all physical features of the proposed project activity in the registered VCS PD /3/ are in place and the PP has operated the project as per registered VCS PD /3/. The project activity consists of a biogas collection system, a biogas treatment system, an electricity generation system and a grid connection system. The installed capacity of the project is 3 MW, involves the installation of 3 sets of gas generators with a unit capacity of 1,000 kW. The electricity generated is transmitted to the local Power Grid via a newly built transformer station, which was then exported to the CCPG. The project activity was expected to supply 11,847 MWh of electricity to the grid annually. The construction of the project started on 15/09/2021, the project has been put into operation on 21/01/2022 verified by checking information on the VCS website and site visit. There are no changes on the key equipment and technology since the validation of the project. No special event which would affect the monitoring of the project has been observed during the monitoring period. The project was under smooth operation except regular repair and maintenance confirmed by checking operation log.

The technical parameters have been verified with the nameplates /9 / as below:

Main Equipment	Parameter	Value
Gas engine	Type	J320GS
	Manufacturer	INNIO Jenbacher GmbH & Co OG
	Rated capacity	1,000 kW

	Rated rotation speed	1,500 r/min
	Lifetime	20 years

The assessment team confirmed that there is no proposed or actual change to the project design during this monitoring period.

All required equipments and procedures are available and implemented in an appropriate manner.

All necessary monitoring instruments are installed. All required instruments including standby and operating procedures for the same have been implemented in an appropriate manner.

The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD /3/. All parameters were monitored and determined as per the monitoring plan which is listed in below table:

Data / Parameter:	$Q_{ww,i,y}$														
Data unit:	m ³ /month														
Description:	The flow of wastewater														
Purpose of the data:	Calculation of baseline emissions Calculation of project emissions														
Parameter value:	610,340.53														
Source of data used:	The flow of wastewater is all sourced from Monthly Report /10/ issued by the project owner.														
Information flow:	2 flow meters installed at project site was measured continuously, recorded hourly and archived electronically. At 24:00 hr of last day of each month, the staff from project owner will read the value for each month and fill out Monthly Report. The data for Monthly Report have been sent to the CDM consulting company for reporting of GHG emission reduction.														
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded hourly by 2 flow meters installed at project site. See below for the information of 2 flow meters verified by site visit and checking calibration certificates /12/: <table border="1"> <thead> <tr> <th>Equipment</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr> </thead> <tbody> <tr> <td>Flow meter F1</td><td>PROMAG 10</td><td>R70DB318000</td><td>0.5</td></tr> <tr> <td>Flow meter F2</td><td>PROMAG 10</td><td>S9040018000</td><td>0.5</td></tr> </tbody> </table>			Equipment	Type	Serial Number	Accuracy	Flow meter F1	PROMAG 10	R70DB318000	0.5	Flow meter F2	PROMAG 10	S9040018000	0.5
Equipment	Type	Serial Number	Accuracy												
Flow meter F1	PROMAG 10	R70DB318000	0.5												
Flow meter F2	PROMAG 10	S9040018000	0.5												

	The type, serial number and accuracy have been confirmed by site visit.		
Calibration:	The calibration information are shown as below /12/:		
	Equipment	Calibration date	Valid until
	Flow meter F1	28/10/2021	27/10/2022
		27/10/2022	26/10/2023
		26/10/2023	25/10/2024
	Flow meter F2	28/10/2021	27/10/2022
		27/10/2022	26/10/2023
		26/10/2023	25/10/2024
The calibration was conducted by accredited third parties which are Guangdong Zhongzhun Test Co., Ltd. /13/ which was accredited by China National Accreditation Service for Conformity Assessment (CNAS).			
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.		
Means of verification:	Data of the parameter was verified by checking Monthly Report. All data is in line with Monthly Report;		
	Information flow was verified by checking Monthly Report, and all information are consistent;		
	Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD;		
	Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.		

Data / Parameter:	BG _{burnt,y}
Data unit:	Nm ³
Description:	Biogas volume in year y
Purpose of the data:	Calculation of baseline emissions Calculation of project emissions
Parameter value:	11,902,032.00
Source of data used:	Biogas volume in year y is all sourced from Monthly Report /10/ issued by the project owner.

Information flow:	1 flow meter installed at project site was measured continuously, recorded hourly and archived electronically. Moreover, the value will be converted automatically into standard value (at the normal condition of 0 °C and 101,325 Pa) as flow meter automatically measure temperature and pressure, expressing LFG volumes in Normalized cubic meters (Nm³). At 24:00 hr of last day of each month, the staff from project owner will read the value for each month and fill out Monthly Report. The data for Monthly Report have been sent to the CDM consulting company for reporting of GHG emission reduction.												
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded hourly by 1 flow meter installed at project site. See below for the information of 1 flow meter verified by site visit and checking calibration certificates /12/: <table><tr><th>Equipment</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>Flow meter F3</td><td>Prosonic Flow B</td><td>S91A7C02000</td><td>1.5</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.			Equipment	Type	Serial Number	Accuracy	Flow meter F3	Prosonic Flow B	S91A7C02000	1.5		
Equipment	Type	Serial Number	Accuracy										
Flow meter F3	Prosonic Flow B	S91A7C02000	1.5										
Calibration:	The calibration information are shown as below /12/: <table><tr><th>Equipment</th><th>Calibration date</th><th>Valid until</th></tr><tr><td rowspan="3">Flow meter F3</td><td>13/09/2021</td><td>12/09/2022</td></tr><tr><td>12/09/2022</td><td>11/09/2023</td></tr><tr><td>11/09/2023</td><td>10/09/2024</td></tr></table> The calibration was conducted by accredited third parties which are Guangdong Zhongzhun Test Co., Ltd. /13/ which was accredited by China National Accreditation Service for Conformity Assessment (CNAS).			Equipment	Calibration date	Valid until	Flow meter F3	13/09/2021	12/09/2022	12/09/2022	11/09/2023	11/09/2023	10/09/2024
Equipment	Calibration date	Valid until											
Flow meter F3	13/09/2021	12/09/2022											
	12/09/2022	11/09/2023											
	11/09/2023	10/09/2024											
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.												
Means of verification:	Data of the parameter was verified by checking Monthly Report. All data is in line with Monthly Report; Information flow was verified by checking Monthly Report, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD; Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.												

Data / Parameter:	WCH4,y												
Data unit:	%												
Description:	Methane content in biogas in the year y												
Purpose of the data:	Calculation of baseline emissions Calculation of project emissions												
Parameter value:	65.19 (average)												
Source of data used:	Methane content in biogas in the year y are all sourced from Monthly Report /10/ issued by the project owner.												
Information flow:	1 gas analyser installed at project site was measured continuously and archived electronically. At 24:00 hr of last day of each month, the monitoring system will automatically calculate the average value of wCH4,y of each month and the staff from project owner read the average value and fill out Monthly Report. The data for Monthly Report have been sent to the CDM consulting company for reporting of GHG emission reduction.												
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded hourly by 1 gas analyser installed at project site. See below for the information of 1 gas analyser verified by site visit and checking calibration certificates /12/: <table><tr><th>Equipment</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>Gas analyser</td><td>BIOGAS 5000</td><td>G502590</td><td>0.5%</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.			Equipment	Type	Serial Number	Accuracy	Gas analyser	BIOGAS 5000	G502590	0.5%		
Equipment	Type	Serial Number	Accuracy										
Gas analyser	BIOGAS 5000	G502590	0.5%										
Calibration:	The calibration information are shown as below /12/: <table><tr><th>Equipment</th><th>Calibration date</th><th>Valid until</th></tr><tr><td rowspan="3">Gas analyser</td><td>09/11/2021</td><td>08/11/2022</td></tr><tr><td>08/11/2022</td><td>07/11/2023</td></tr><tr><td>07/11/2023</td><td>06/11/2024</td></tr></table> The calibration was conducted by accredited third parties which are Shenzhen Academy of Metrology & Quality Inspection National Hi-tech Metrology Station /13/ which was accredited by Guangdong Market Supervision Administration.			Equipment	Calibration date	Valid until	Gas analyser	09/11/2021	08/11/2022	08/11/2022	07/11/2023	07/11/2023	06/11/2024
Equipment	Calibration date	Valid until											
Gas analyser	09/11/2021	08/11/2022											
	08/11/2022	07/11/2023											
	07/11/2023	06/11/2024											

QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Data of the parameter was verified by checking Monthly Report. All data is in line with Monthly Report; Information flow was verified by checking Monthly Report, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD; Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.

Data / Parameter:	COD _{ww,untreated,y}
Data unit:	tCOD/m ³
Description:	The chemical oxygen demand of the wastewater before the treatment system affected by the project activity
Purpose of the data:	Calculation of baseline emissions Calculation of project emissions
Parameter value:	0.057818 (average)
Source of data used:	The chemical oxygen demand of the wastewater before the treatment system affected by the project activity are all sourced from Monthly Report /10/ issued by the project owner.
Information flow:	1 spectrophotometer installed at laboratory of project site was measured and record on a daily basis. Each day, the staff of project would collect the sample from waste water before the treatment system and sent to laboratory of project for test by using spectrophotometer based on the requirement of Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017). The test result would be recorded in the Monthly Report. The confidence/precision level of 90/10 would be assured. The data for Monthly Report have been sent to the CDM consulting company for reporting of GHG emission reduction.
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded hourly by 1 spectrophotometer installed at laboratory of project site. See below for the information of 1 spectrophotometer verified by site visit and checking calibration certificates /12/:

	Equipment	Type	Serial Number	Accuracy
	Spectrophotometer	UV756	YD03041905012	0.5 nm
The type, serial number and accuracy have been confirmed by site visit.				
Calibration:	The calibration information are shown as below /12/:			
	Equipment	Calibration date	Valid until	
	Spectrophotometer	28/10/2021	27/10/2022	
		27/10/2022	26/10/2023	
		26/10/2023	25/10/2024	
	The calibration was conducted by accredited third parties which are Guangdong Zhongzhun Test Co., Ltd. /13/ which was accredited by China National Accreditation Service for Conformity Assessment (CNAS).			
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.			
Means of verification:	Data of the parameter was verified by checking Monthly Report. All data is in line with Monthly Report;			
	Information flow was verified by checking Monthly Report, and all information are consistent;			
	Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD;			
	Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.			

Data / Parameter:	COD _{ww,treated,y}
Data unit:	tCOD/m ³
Description:	The chemical oxygen demand of the wastewater after the treatment system affected by the project activity
Purpose of the data:	Calculation of baseline emissions Calculation of project emissions
Parameter value:	0.014907 (average)
Source of data used:	The chemical oxygen demand of the wastewater after the treatment system affected by the project activity are all sourced from Monthly Report /10/ issued by the project owner.

Information flow:	1 spectrophotometer installed at laboratory of project site was measured and record on a daily basis. Each day, the staff of project would collect the sample from waste water after the treatment system and sent to laboratory of project for test by using spectrophotometer based on the requirement of Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017). The test result would be recorded in the Monthly Report. The confidence/precision level of 90/10 would be assured. The data for Monthly Report have been sent to the CDM consulting company for reporting of GHG emission reduction.			
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded hourly by 1 spectrophotometer installed at laboratory of project site. See below for the information of 1 spectrophotometer verified by site visit and checking calibration certificates /12/:			
	Equipment	Type	Serial Number	Accuracy
	Spectrophotometer	UV756	YD03041905012	0.5 nm
	The type, serial number and accuracy have been confirmed by site visit.			
Calibration:	The calibration information are shown as below /12/:			
	Equipment	Calibration date	Valid until	
	Spectrophotometer	28/10/2021	27/10/2022	
		27/10/2022	26/10/2023	
		26/10/2023	25/10/2024	
	The calibration was conducted by accredited third parties which are Guangdong Zhongzhun Test Co., Ltd. /13/ which was accredited by China National Accreditation Service for Conformity Assessment (CNAS).			
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.			
Means of verification:	Data of the parameter was verified by checking Monthly Report. All data is in line with Monthly Report; Information flow was verified by checking Monthly Report, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD; Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.			

Data / Parameter:	EGPJ,facility,y												
Data unit:	MWh												
Description:	Quantity of net electricity generated supplied by the project plant/unit in year y												
Purpose of the data:	Calculation of baseline emissions												
Parameter value:	20,740.137												
Source of data used:	Quantity of net electricity generated supplied by the project plant/unit in year y are all sourced from Monthly Report /10/ issued by the project owner, ETNs /11/ issued by power grid company covering monitoring period.												
Information flow:	1 electricity meter installed at project site (E) was measured continuously, recorded monthly and archived electronically. At 24:00 hr of last day of each month, the staff from project owner will record 1 electricity meters' readings and fill out Monthly Report. The staff from power grid company will record the meter readings of meter (E) then transcribes the data into Electricity Transaction Notes (ETNs), then after the confirmation of the project owner for the ETNs, the project owner will issue the invoice as sales receipts. The data for Monthly Report and ETNs have been sent to the CDM consulting company for reporting of GHG emission reduction. The conservative one would be used for ER calculation. All ETNs has been checked and data from ETNs has been crosschecked with the same form monthly report.												
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 1 electricity meter installed at project site. See below for the information of 1 electricity meter verified by site visit and checking calibration certificates /12/: <table><tr><th>Meter</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>E</td><td>DTS858</td><td>RM221100098258</td><td>0.2s</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.			Meter	Type	Serial Number	Accuracy	E	DTS858	RM221100098258	0.2s		
Meter	Type	Serial Number	Accuracy										
E	DTS858	RM221100098258	0.2s										
Calibration:	The calibration information are shown as below /12/: <table><tr><th>Meter</th><th>Calibration date</th><th>Valid until</th></tr><tr><td rowspan="3">E</td><td>31/08/2021</td><td>30/08/2022</td></tr><tr><td>30/08/2022</td><td>29/08/2023</td></tr><tr><td>29/08/2023</td><td>28/08/2024</td></tr></table>			Meter	Calibration date	Valid until	E	31/08/2021	30/08/2022	30/08/2022	29/08/2023	29/08/2023	28/08/2024
Meter	Calibration date	Valid until											
E	31/08/2021	30/08/2022											
	30/08/2022	29/08/2023											
	29/08/2023	28/08/2024											

	The calibration was conducted by accredited third parties which are Metrology Center of EPH /13/ which was accredited by Henan Market Supervision Administration.
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Data of the parameter was verified by checking Monthly Report and ETNs. All data is in line with Monthly Report and ETNs; Information flow was verified by checking Monthly Report and ETNs, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD; Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.

Data / Parameter:	EC _{PJ,y}
Data unit:	MWh
Description:	Amount of electricity consumed by the project activity in year y
Purpose of the data:	Calculation of project emissions
Parameter value:	2.388
Source of data used:	Amount of electricity consumed by the project activity in year y are all sourced from Monthly Report /10/ issued by the project owner, ETNs /11/ issued by power grid company covering monitoring period.
Information flow:	1 electricity meter installed at project site (E) was measured continuously, recorded monthly and archived electronically. At 24:00 hr of last day of each month, the staff from project owner will record 1 electricity meters' readings and fill out Monthly Report. The staff from power grid company will record the meter readings of meter (E) then transcribes the data into Electricity Transaction Notes (ETNs), then after the confirmation of the project owner for the ETNs, the power grid company will issue the invoice as sales receipts. The data for Monthly Report and ETNs have been sent to the CDM consulting company for reporting of GHG emission reduction. The conservative one would be used for ER calculation. All ETNs has been checked and data from ETNs has been crosschecked with the same form monthly report.

Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 1 electricity meter installed at project site. See below for the information of 1 electricity meter verified by site visit and checking calibration certificates /12/:			
	Meter	Type	Serial Number	Accuracy
	E	DTS858	RM221100098258	0.2s
	The type, serial number and accuracy have been confirmed by site visit.			
Calibration:	The calibration information are shown as below /12/:			
	Meter	Calibration date	Valid until	
	E	31/08/2021	30/08/2022	
		30/08/2022	29/08/2023	
		29/08/2023	28/08/2024	
The calibration was conducted by accredited third parties which are Metrology Center of EPH /13/ which was accredited by Henan Market Supervision Administration.				
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.			
Means of verification:	Data of the parameter was verified by checking Monthly Report and ETNs. All data is in line with Monthly Report and ETNs;			
	Information flow was verified by checking Monthly Report and ETNs, and all information are consistent;			
	Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PD;			
	Calibration was verified by checking calibration certificate, all calibration of monitoring equipment meets the requirement indicated in the PD.			

Data / Parameter:	P
Data unit:	Pa
Description:	Pressure of the biogas
Purpose of the data:	Calculation of baseline emissions
Parameter value:	/
Source of data used:	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and

	temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas.
Information flow:	/
Monitoring method, frequency and equipments:	/
Calibration:	/
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	/

Data / Parameter:	T
Data unit:	°C
Description:	Temperature of the biogas
Purpose of the data:	Calculation of baseline emissions
Parameter value:	/
Source of data used:	The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas.
Information flow:	/
Monitoring method, frequency and equipments:	/
Calibration:	/
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	/

Parameters available at validation stage:

Below data has been verified against the data sources and the PD, for the rest of parameters are not used in the calculation of ER in this monitoring period.

Parameter title	Description	Data	Source
$MCF_{ww,treatment,BL,i}$	Methane correction factor for baseline wastewater treatment systems i	0.8	AMS-III.H /7/
$\eta_{COD,BL,i}$	COD removal efficiency of the baseline treatment system	76.33%	Operation log of waste treatment /29/ (this value was not used in the ER calculation as update value was used based on ex-post monitoring parameter)
GWP_{CH_4}	Global warming potential of methane	28	IPCC /15/
$B_{o,ww}$	Methane producing capacity of the wastewater	0.25	AMS-III.H /7/
UF_{BL}	Model correction factor to account for model uncertainties	0.89	AMS-III.H /7/
UF_{PJ}	Model correction factor to account for model uncertainties	1.12	AMS-III.H /7/
$MCF_{ww,treatment,PJ,k}$	Methane correction factor for project wastewater treatment system k	0.8	AMS-III.H /7/
CFE_{ww}	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems	0.9	AMS-III.H /7/

D _{CH4}	Methane density	0.0007156	Tool to determine the mass flow of a greenhouse gas in a gaseous stream (version 03.0) /30/
EF _{grid,OM,y}	Operation margin emission factor of CCPG	0.8587	2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA /14/
EF _{grid,BM,y}	Build margin emission factor of CCPG	0.2854	2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA /14/
TDL _{j,y}	Average technical transmission and distribution losses for providing electricity to source j, k in year y	20	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0) /31/
FE	Flare efficiency in year y	100	AMS-III.H /7/

In conclusion, the assessment team confirmed GHG emission reductions and removals have been quantified correctly in accordance with the project description and applied methodology.

4.4 Quality of Evidence to Determine Reductions and Removals

The monitoring has been carried out in accordance with VCS PD /3/ dated 25/06/2023.

As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the MR Report and ER Calculation Spreadsheet /2/ and are consistent with the applied methodology AMS-III.H version 19.0 and the monitoring plan. The parameters are complete in this monitoring period.

By checking the original record, crosscheck with the supporting evidence issued from other party than project owner, checking calibration related documents and interview with project owner through the site visit, the VVB is able to confirm there are no transposition errors between data sets. All data are consistent in all data sets.

After verifying the reported figures with the raw data sources, it's confirmed that the values of the parameters from the raw data sources are consistent with those quoted in the ER Calculation

Spreadsheet and the MR Report. The verification process for the same has been clearly described above in section 4.3 of the report.

4.5 Non-Permanence Risk Analysis

Not applicable as a biogas project.

5 VERIFICATION OPINION

5.1 Verification Summary

LGAI Technological Center S.A. (hereinafter referred to as Applus+ Certification) has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the verification of greenhouse gas emission reductions of the project activity “Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project” (VCS Ref. No. 3851).

The management of Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd. is responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the VCS PD /3/ dated 25/06/2023.

Our verification approach was based on the requirements as defined under the applicable VCS standards, relevant UNFCCC requirements and ISO 14064-3: 2019. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered PD;
- the monitoring plan in the registered PD is as per the applied methodology;
- the monitoring complies with the registered PD;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

In our opinion, the GHG emission reductions for “Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project” during the monitoring period 21/01/2022 to 31/12/2023 as reported in Monitoring Report, prepared on the basis of the project’s Monitoring Plan are fairly stated.

5.2 Verification Conclusion

Our verification approach was based on the requirements as defined under the applicable VCS standards, relevant UNFCCC requirements and ISO 14064-3: 2019. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The achieved level of assurance for this verification is thus a reasonable level of

assurance as per the requirements set out in the applied VCS Standard version 4.7 clause 4.1.10 (1). Based on the information we have seen and evaluated, we confirm the following statement:

Verification period: From 21/01/2022 to 31/12/2023

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU's (tCO ₂ e)	Removal VCU's (tCO ₂ e)	Total VCUs (tCO ₂ e)
21/01/2022 to 31/12/2022	66,334	8,738	0	57,596	0	57,596
01/01/2023 to 31/12/2023	75,253	9,227	0	66,026	0	66,026
Total	141,587	17,965	0	123,622	0	123,622

5.3 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
21/01/2022 to 31/12/2022	62,677	57,596	-8.11%	Conservative as less ER achieved also the difference to mainly due to following reason: (1) Less net electricity supplied to the grid generated; (2) Insufficient management experience due to the 1 st year operation.
01/01/2023 to 31/12/2023	66,310	66,026	-0.43%	Conservative as less ER achieved, also consider this is under normal variation due to lots of factors could influence the ER.
Total	128,987	123,622	-4.16%	Conservative as less ER achieved, also consider this is under normal variation due to lots of factors could influence the ER.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

<i>Section</i>	<i>Information</i>	<i>Justification</i>	<i>Assessment method and conclusion</i>
<u>Not applicable</u>	<u>Not applicable</u>	<u>Not applicable</u>	<u>Not applicable</u>

APPENDIX 2: REFERENCE LIST

1. Monitoring report, Version 01, dated 22/03/2024; Version 03, dated 19/06/2024
2. ER calculation spreadsheet
3. VCS PD, version 04, dated 25/06/2023
4. VCS Validation report, No. A+SH_SYST_VCS_VAL_11522, version 01.3, completed by Applus+ Certification
5. VCS standard version 4.7, dated on 16/04/2024
6. Statement issued by project owner
7. Approved methodology AMS-III.H: Methane recovery in wastewater treatment, version 19.0;
AMS-I.D: Grid connected renewable electricity generation, version 18.0
8. CDM Monitoring procedure
9. Nameplate of the equipment
10. Monthly Report
11. Electricity Transaction Notes covering the monitoring period
12. Calibration certificates of electricity meter E covering the whole monitoring period issued by Metrology Center of EPH

Calibration certificates of flow meter F1, F2 and F3 covering the whole monitoring period issued by Guangdong Zhongzhun Test Co., Ltd.

Calibration certificates of gas analyser covering the whole monitoring period issued by Shenzhen Academy of Metrology & Quality Inspection National Hi-tech Metrology Station

Calibration certificates of spectrophotometer covering the whole monitoring period issued by Guangdong Zhongzhun Test Co., Ltd.

- 13 Accreditation certificates for Metrology Center of EPH issued by Henan Market Supervision Administration

Accreditation certificates for Guangdong Zhongzhun Test Co., Ltd. issued by China National Accreditation Service for Conformity Assessment (CNAS)

Accreditation certificates for Shenzhen Academy of Metrology & Quality Inspection National Hi-tech Metrology Station issued by Guangdong Market Supervision Administration.

- 14 2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA

- 15 IPCC

- 16 Staff roster of project

- 17 Power Purchase Agreement and Invoice

- 18 Emissions from solid waste disposal sites, version 08.0

- 19 Technical literature

- 20 I-REC website: <https://www.trackingstandard.org/>

- 21 Cooperation agreement signed by the project owner and Zhengzhou Dongxing Environmental Protection Energy Co., Ltd

- 22 Grievance expression book

- 23 Questionnaires
- 24 Meeting minutes
- 25 Project approval
- 26 EIA approval
- 27 Regulation and laws:
 - “Women’s Rights Protection Law”
 - “Minor Protection Law”
 - “Labor Law”
 - “Criminal Law”
- 28 EIA report
- 29 Operation log of waste treatment
- 30 Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0
- 31 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
- 32 Test report issued by Henan Provincial Government Testing Co., LTD,
- 33 Hazardous waste collection and disposal contract with Henan Junchi Environmental Protection Technology Co., LTD.

